



UART1 테스트

CRZ Technology

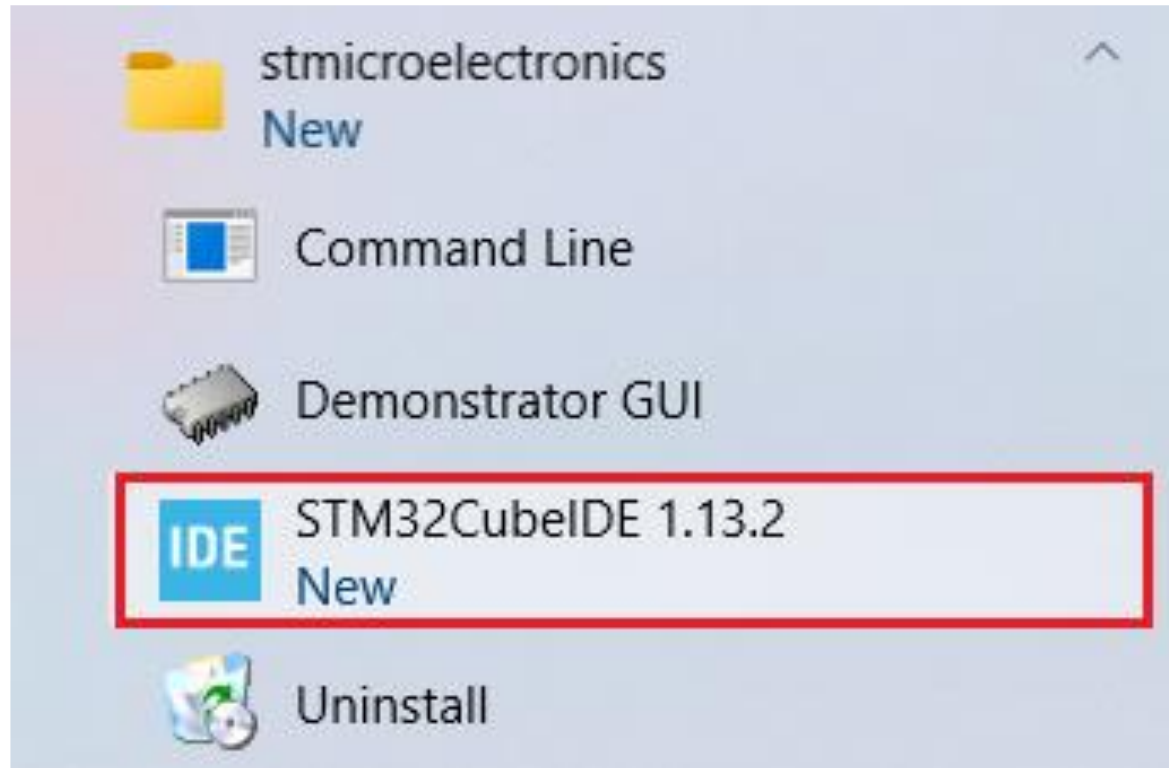




Mango-M32F2 UART1 테스트

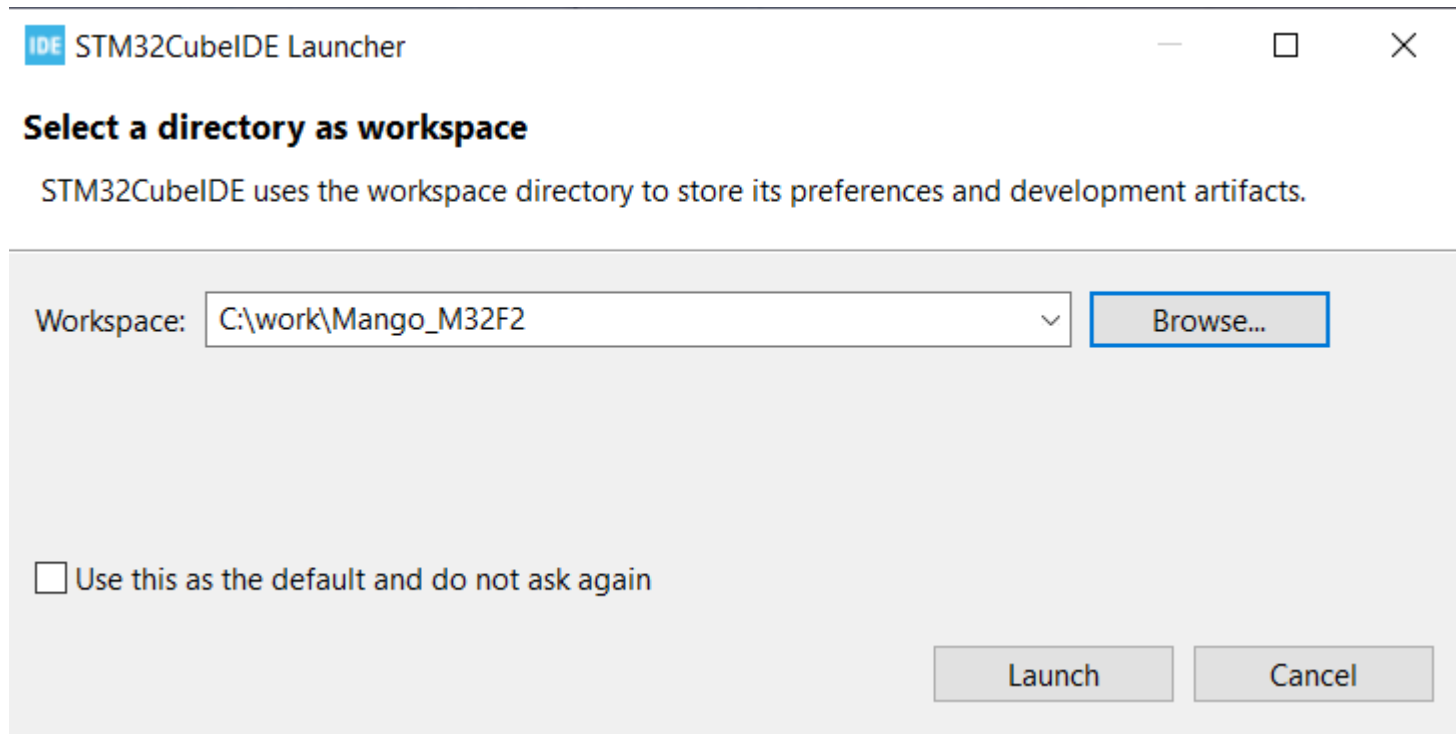
STM32CubeIDE 프로젝트 생성

■ STM32CubeIDE 실행.

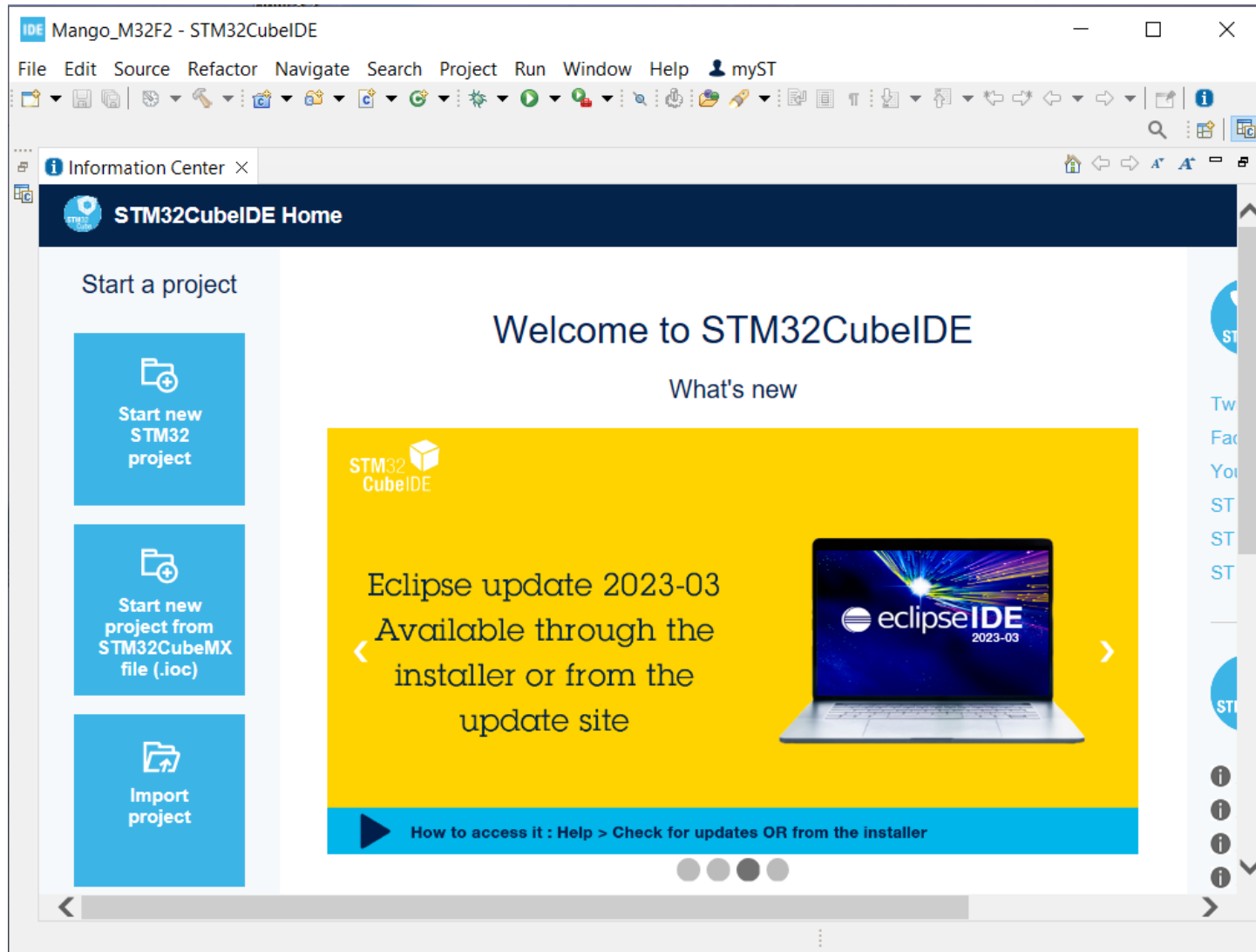


STM32CubeIDE 프로젝트 생성

■ workspace 설정

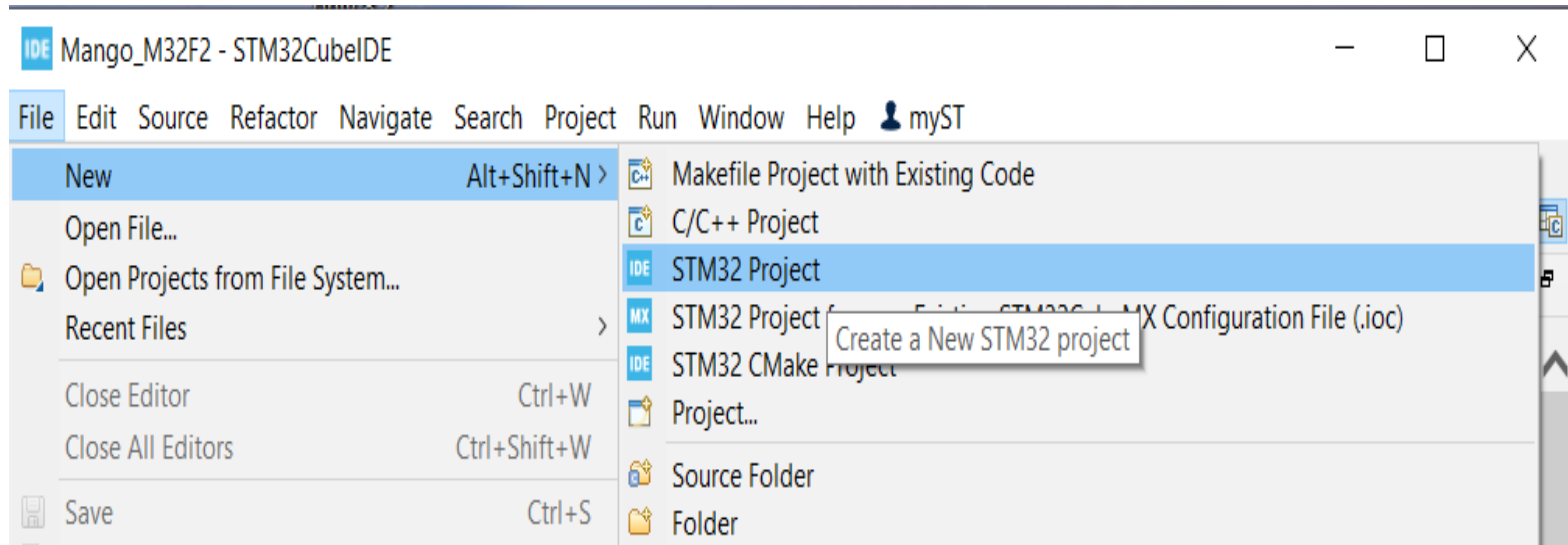


STM32CubeIDE 프로젝트 생성



STM32CubeIDE 프로젝트 생성

- “File” 클릭, “New” 클릭, “STM32 Project” 클릭.



STM32CubeIDE 프로젝트 생성

“STM32F207VGT6” 선택 후 "Next" 클릭

STM32 Project

Target Selection

Select STM32 target or STM32Cube example

MCU/MPU Selector | Board Selector | Example Selector | Cross Selector

MCU/MPU Filters

Commercial Part Number:

PRODUCT INFO

- Segment >
- Series >
- Line >
- Marketing Status >
- Price >
- Package >

STM32F2 Series

STM32F207VGT6

High-performance Arm Cortex-M3 MCU with 1 Mbyte of Flash memory, 120 MHz CPU, ART Accelerator, Ethernet

Unit Price for 10kU (US\$) : 7.6885

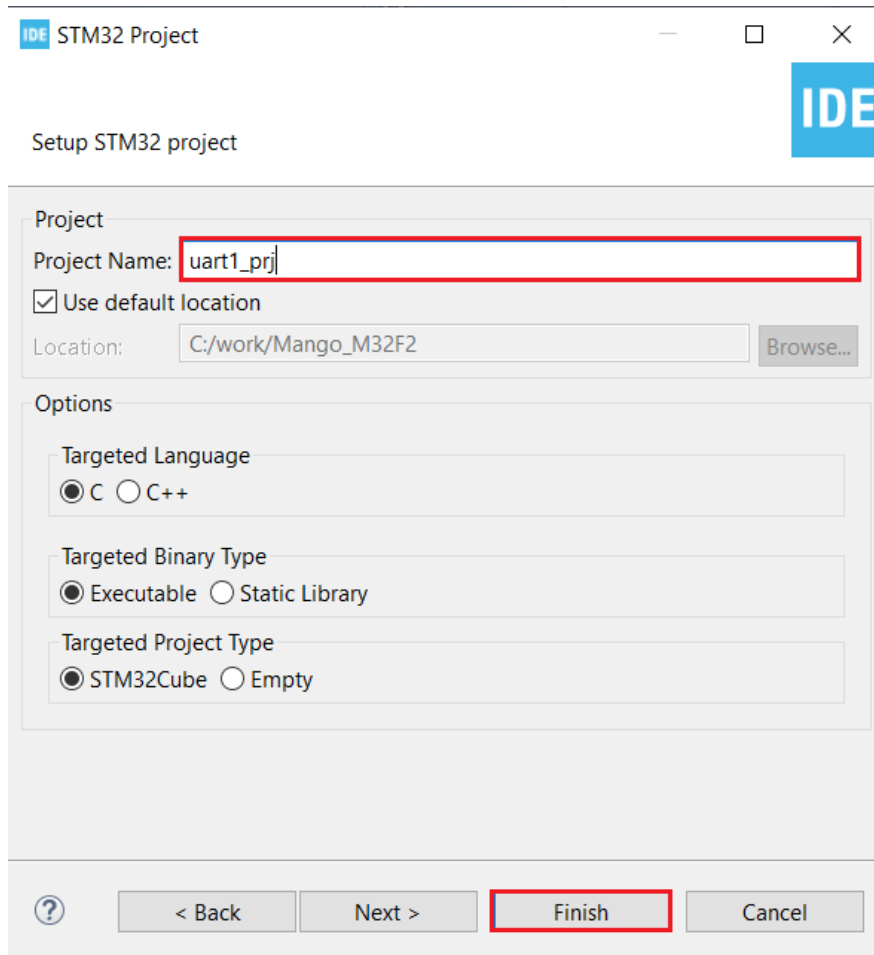
MCUs/MPUs List: 3 items

Commercial Part No	Part No	Reference	Ma...	Package	Fla...	RAM	Fr...
STM32F207VGT6		STM32F207VGTx	Active	LQFP 100 14x...	102...	128 ... 82	120 ...
STM32F207VGT6J	STM32F207VG	STM32F207VGTx	NRND	LQFP 100 14x...	102...	128 ... 82	120 ...
STM32F207VGT6...		STM32F207VGTx	Active	LQFP 100 14x...	102...	128 ... 82	120 ...

Next >

STM32CubeIDE 프로젝트 생성

- Project Name으로 “uart1_prj” 입력 후 “Finish” 클릭.



IDE STM32 Project

Setup STM32 project

Project

Project Name:

☒ Use default location

Location:

Options

Targeted Language

☒ C ☐ C++

Targeted Binary Type

☒ Executable ☐ Static Library

Targeted Project Type

☒ STM32Cube ☐ Empty

Pinout view

Mango_M32F2 - Device Configuration Tool - STM32CubeIDE

File Edit Navigate Search Project Run Window Help Hello

uart1_prj.ioc x

uart1_prj.ioc - Pinout & Configuration

Pinout & Configuration Clock Configuration Project Manager Tools

Software Packs Pinout

Pinout view System view

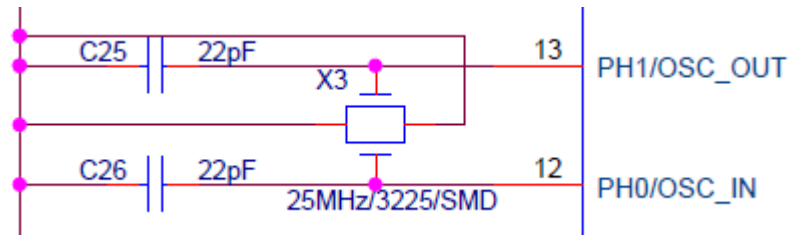
Categories A->Z

- System Core
- Analog
- Timers
- Connectivity
- Multimedia
- Security
- Computing
- Middleware and Software Packs

STM32F207VGTx LQFP100

Clock / UART1 pinmap

■ 25MHz external clock - PH0 / PH1

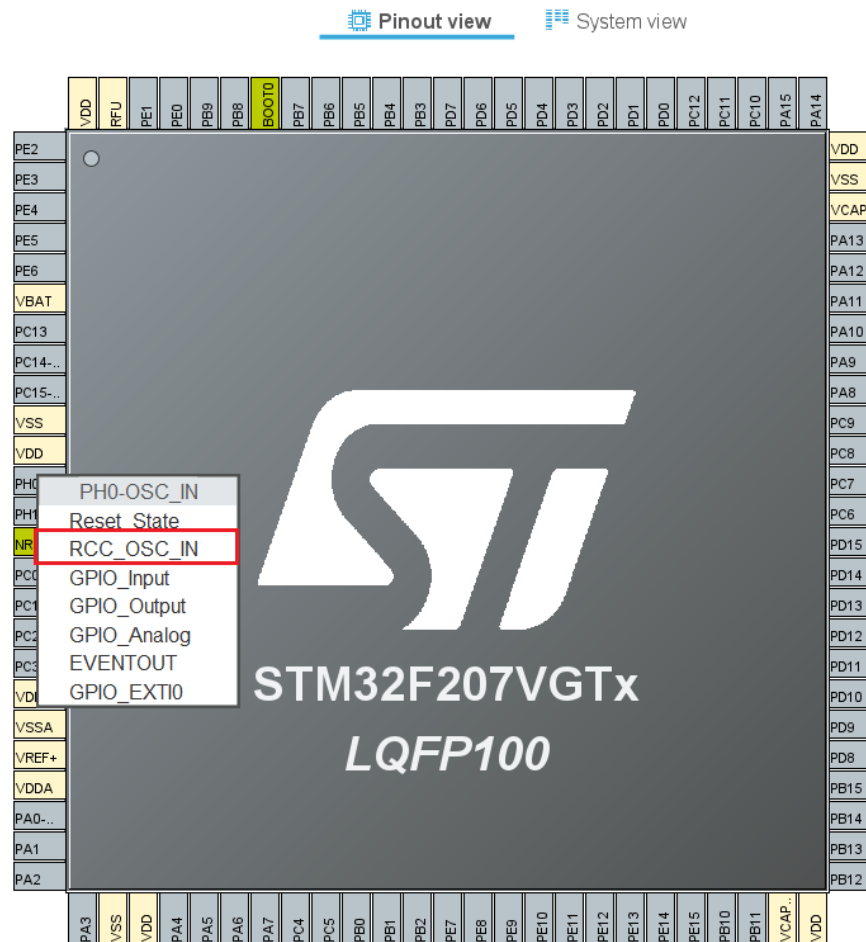


■ UART1 - PB6 / PB7

PB5	91	PB5	UART1_TX
PB6	92	PB6	UART1_RX
PB7	93	PB7	I2C1_SCL
PB8	95	PB8	

Pinout view

■ PH0 – RCC_OSC_IN 선택

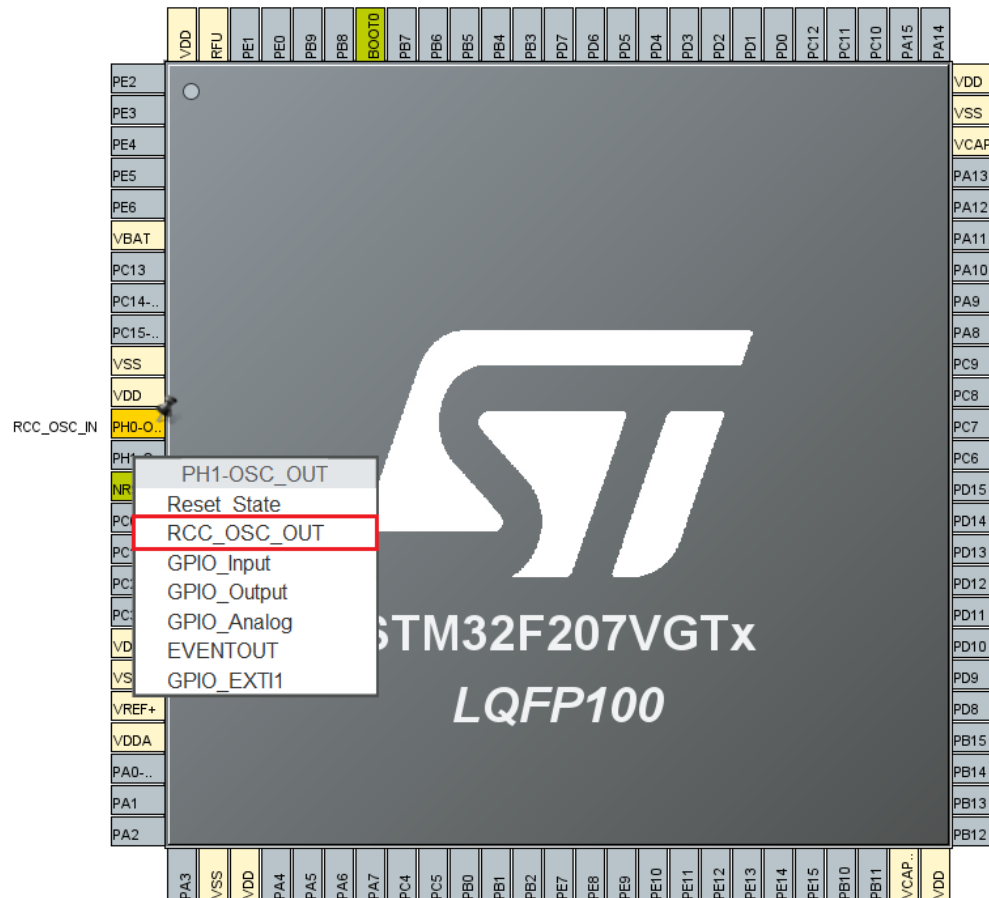


Pinout view

■ PH1 – RCC_OSC_OUT 선택

Pinout view

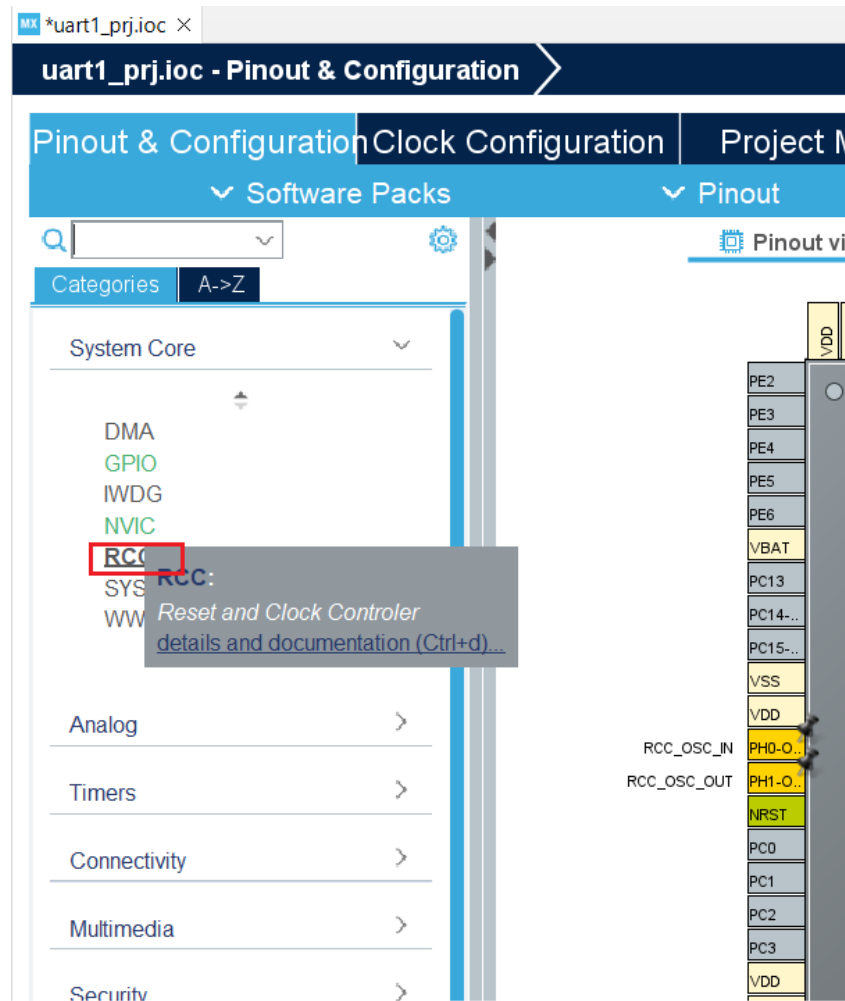
System view





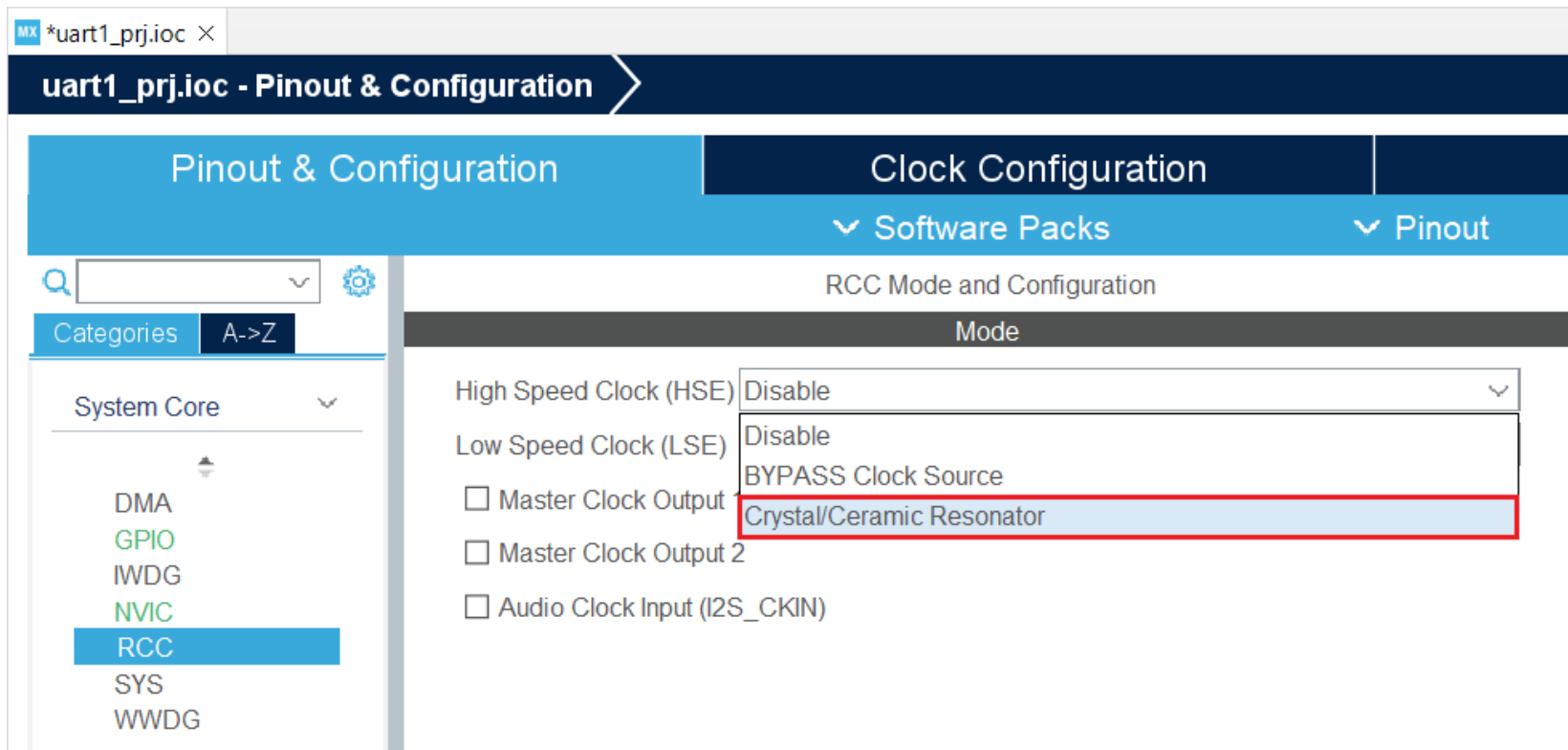
RCC 설정

■ “System Core”에서 “RCC” 클릭.



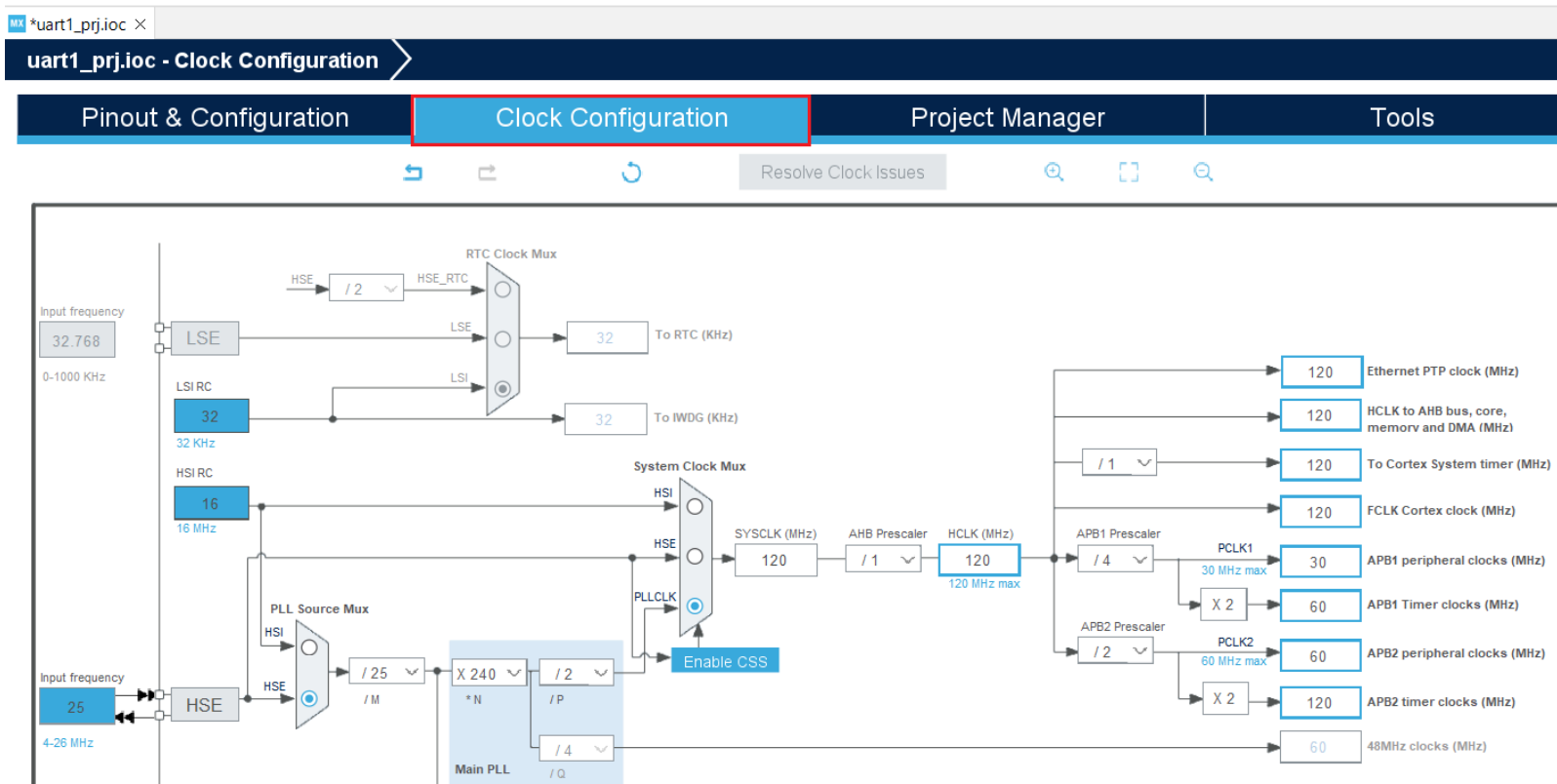
RCC 설정

- “High Speed Clock(HSE)”에서 “Crystal/Ceramic Resonator” 클릭



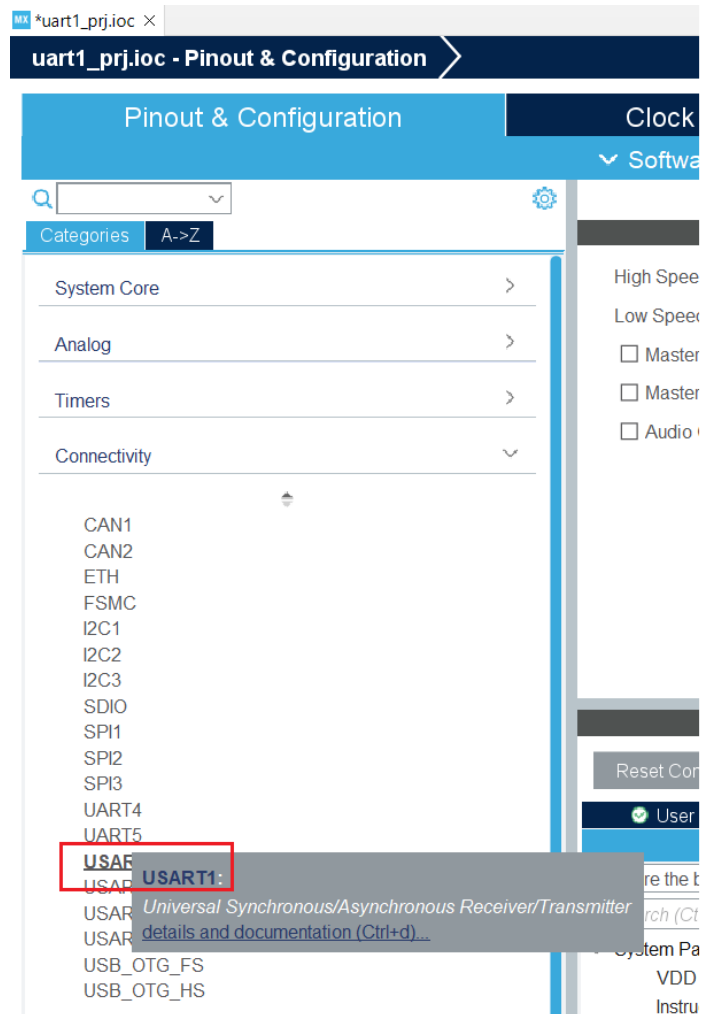
Clock Configuration

■ “Clock Configuration” 클릭 후 Clock Frequency 설정



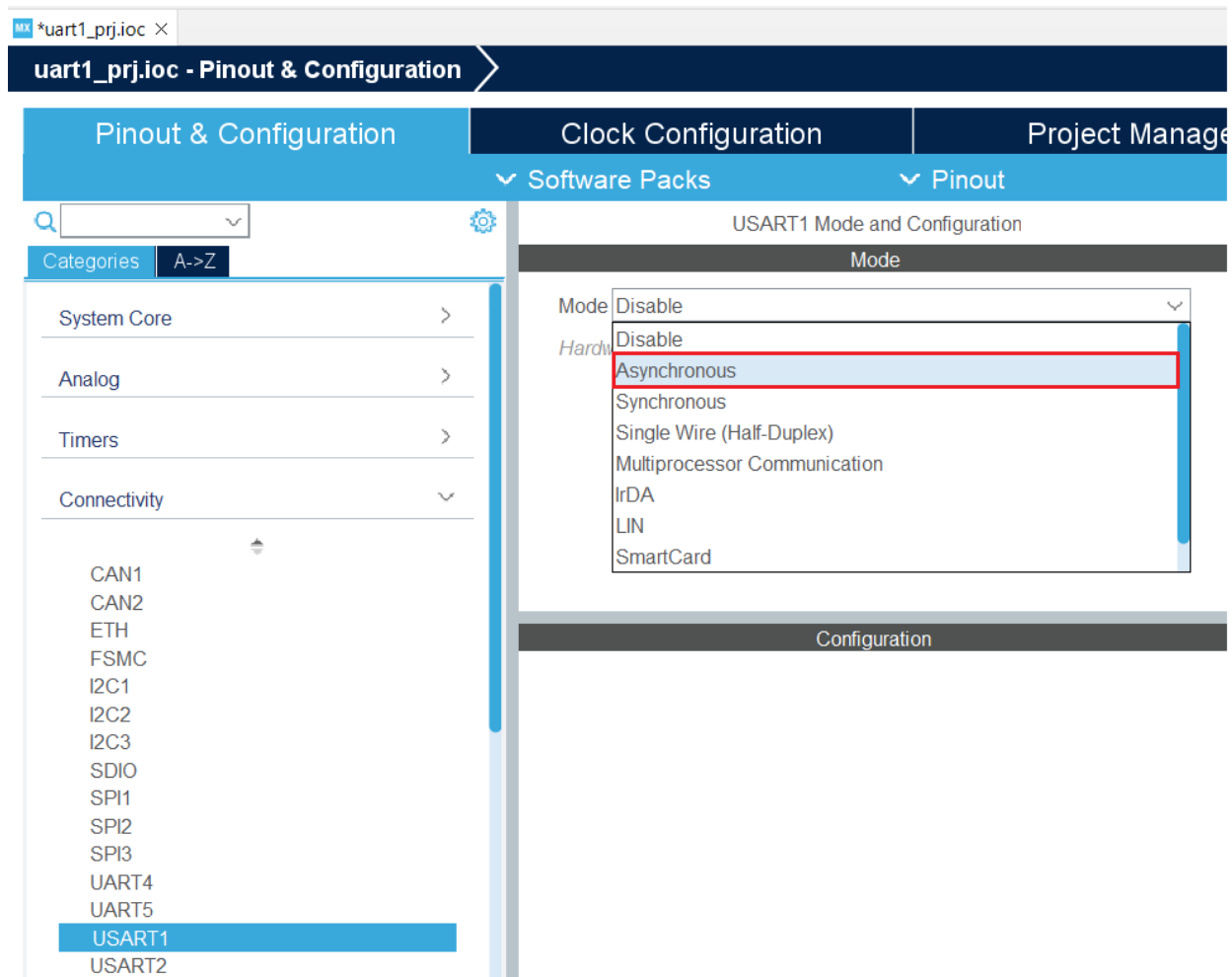
UART1 설정

- “Connectivity” 클릭, “USART1” 클릭.



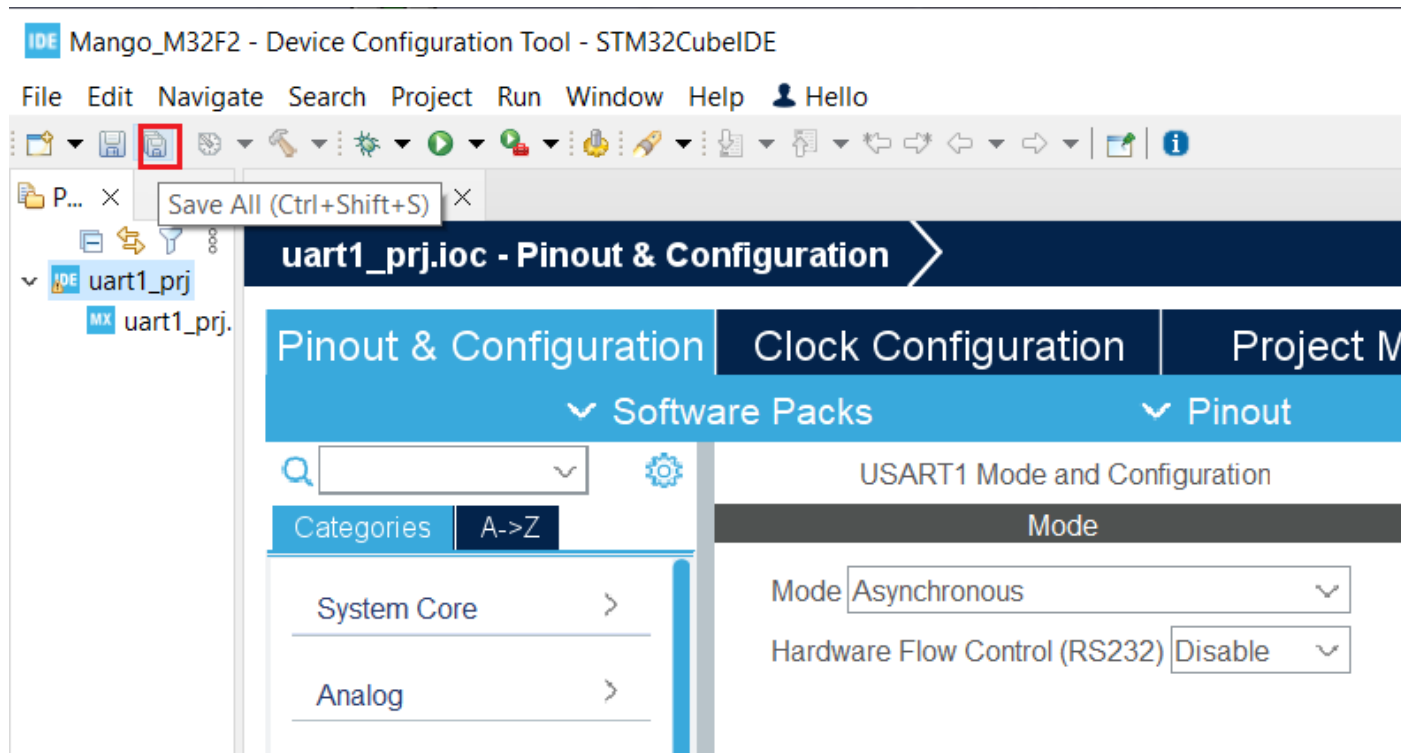
UART1 설정

■ “Mode“, “Asynchronous” 클릭.



UART1 Test Project

- “Save All”클릭, Code Generation.



UART1 Test Project

■ core/src/main.c 편집

- printf() – UART1, overwrite _write() function

```
/* Private user code -----*/
/* USER CODE BEGIN 0 */
#define hUART huart1
int _write( int32_t file , uint8_t *ptr , int32_t len )
{
    /* Implement your write code here, this is used by puts and printf for example */
    for ( int16_t i = 0 ; i < len ; ++i )
    {
        HAL_UART_Transmit( &hUART, ptr++, 1, 100);
    }
    return len;
}
/* USER CODE END 0 */

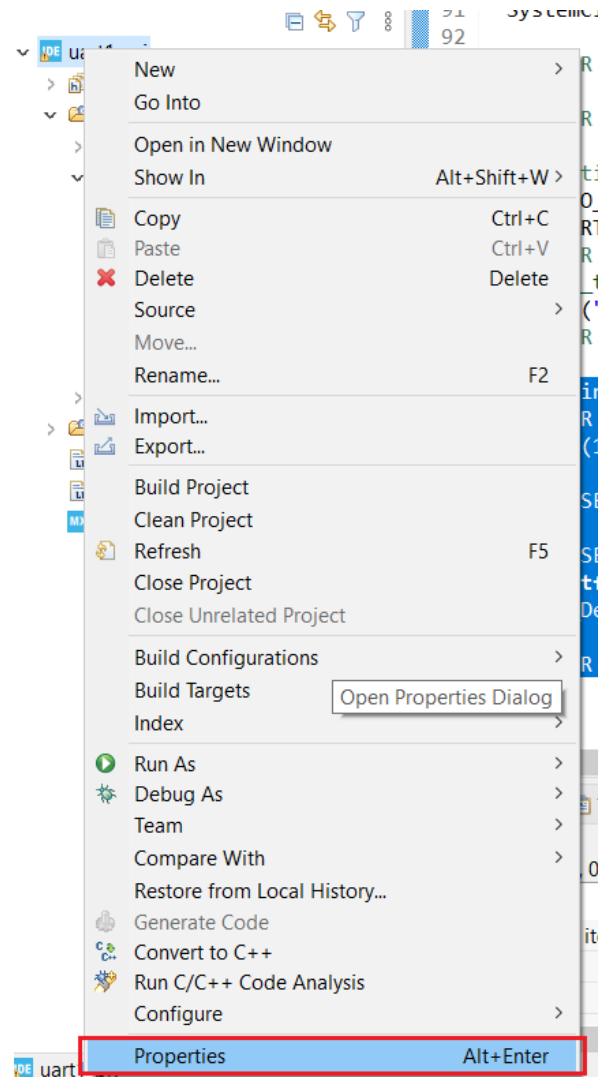
/* USER CODE BEGIN 2 */
int count = 0;
printf("UART1 test\r\n");
/* USER CODE END 2 */

/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
    printf("count=%d\r\n", count++);
    HAL_Delay(1000);
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
}
/* USER CODE END 3 */
```

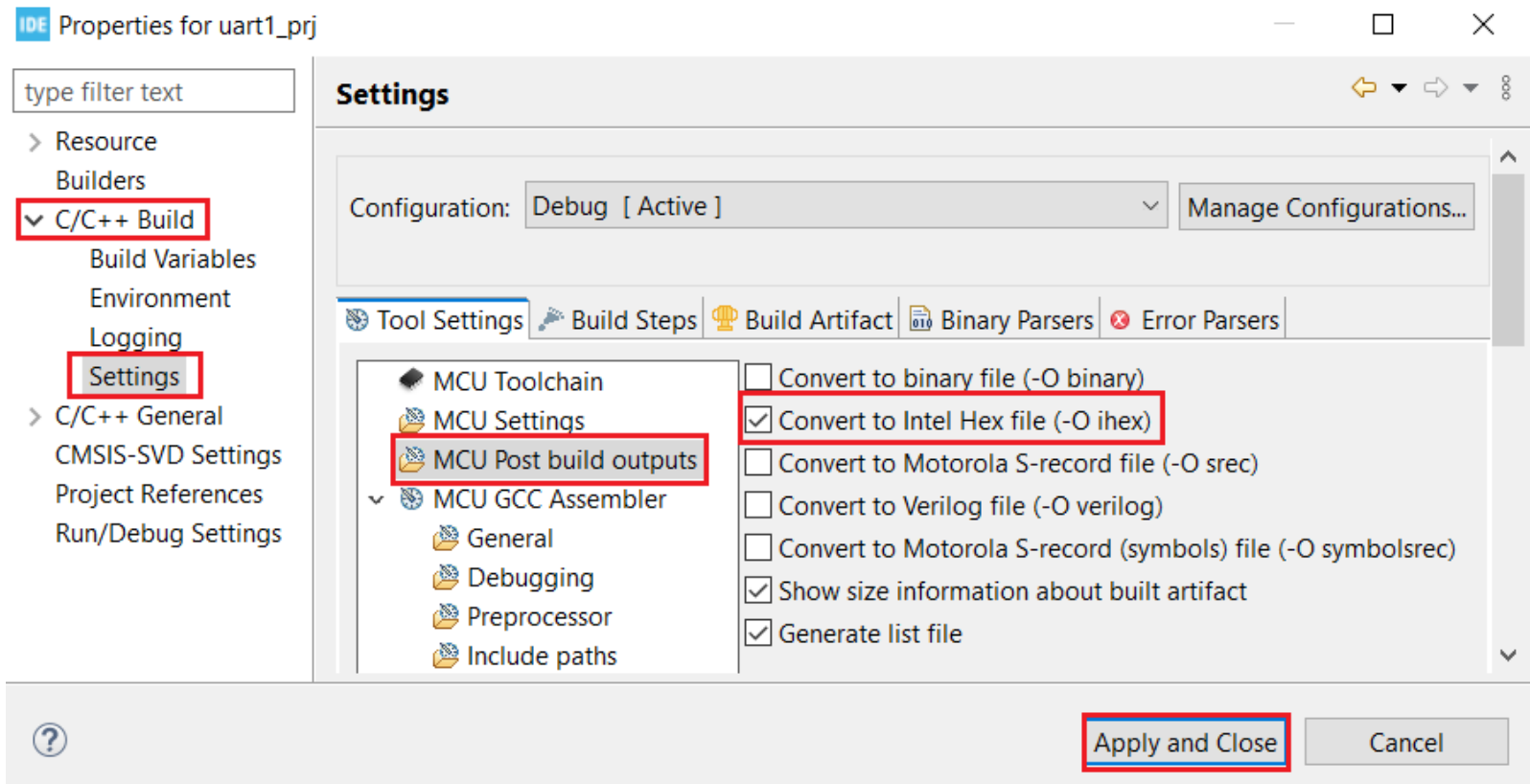
Hex File Generation

- “uart1_prj” 우클릭, “Properties” 클릭.



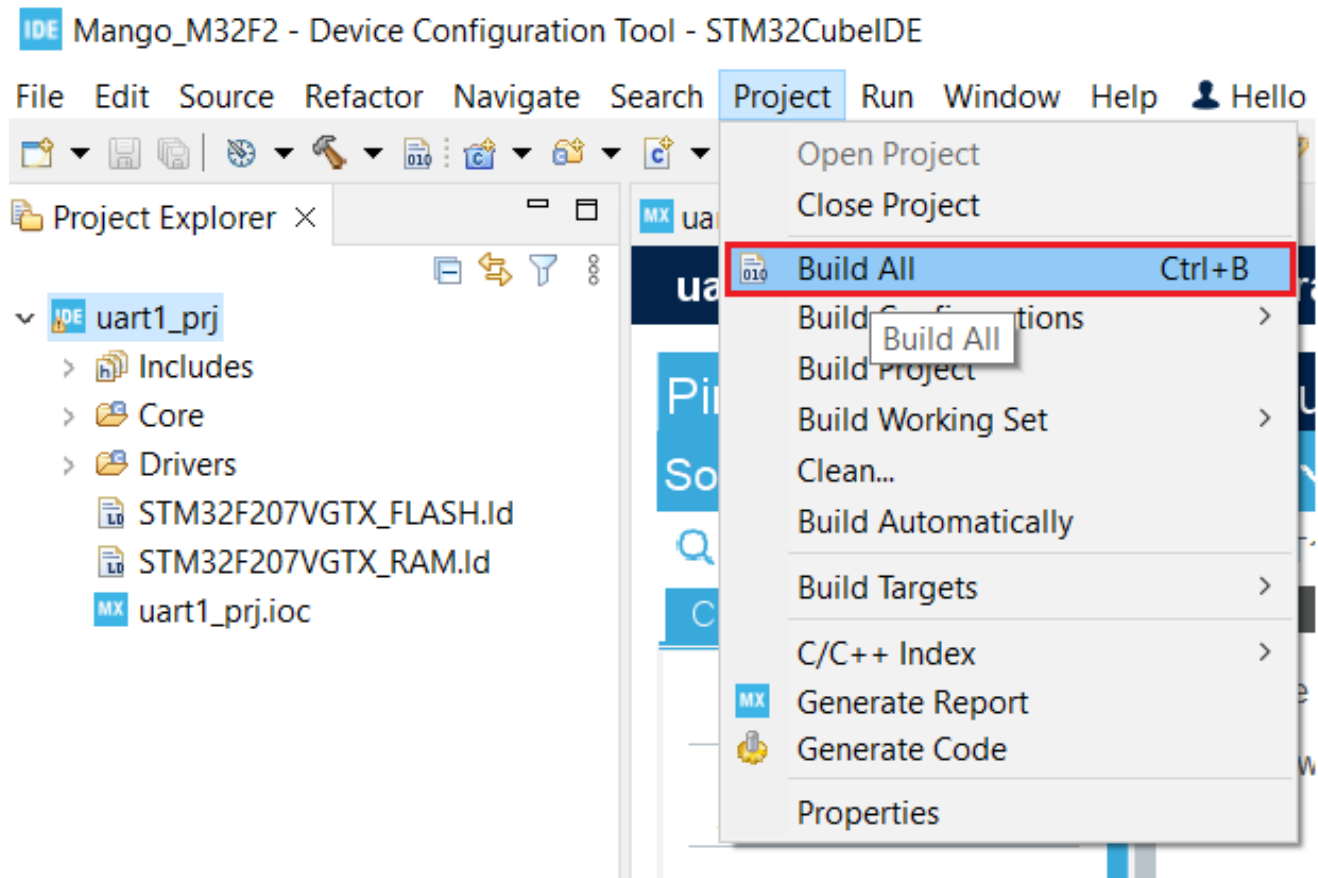
Hex File Generation

- “C/C++ Build”-“Settings”-“Tool Settings”-“MCU Post build outputs”-“Convert to Intel Hex file (-O ihex) 클릭,
- “Apply and Close” 클릭.



Project Build

- “Project”클릭, “Build All” 클릭.



Test 결과 확인

- 3핀 serial cable을 UART3에 연결하고 Flash Loader로 다운로드.
- 3핀 serial cable을 UART1에 연결하고 결과 확인.
- RCC / Clock Configuration / UART1 설정은 나머지 실습에 공통.



A screenshot of a Tera Term VT window titled "COM5:115200baud - Tera Term VT". The window has a menu bar with "File", "Edit", "Setup", "Control", "Window", and "Help". The main text area displays the following output:

```
UART1 test  
count=0  
count=1  
count=2  
count=3  
count=4  
count=5  
count=6  
count=7
```


Thank You

